

Herschel, William **98–99**
 Hertz, Heinrich **150**
 Hewish, Antony 302
 Hieron II of Syracuse 19, 20
 Higgs, Peter **311**
 Hilbert, David 254
 Hildegard of Bingen **37**
 Hipparchus **38**
 Hippocrates **27**
 Hiroshima 235
 histology 164
 Hodgkin, Dorothy **250**
 Hodgkin's lymphoma 303
 Hooke, Robert 68, **70–75**
 Hopper, Grace **242–243**
 hormones
 insulin 224
 synthesis of human 228
 horoscopes 57
 House of Wisdom (Baghdad) 34
 Hoyle, Fred **310**
 HTML 306
 HTTP 306
 Hubble, Edwin **220–223**
 Hubble's law 223
 human ancestry 88, 91, 133, 137
 Human Genome Project 265, 277
 Humboldt, Alexander von **160**
 humors, four 26–27, 37
 Hunter, John 106–108
 Hutton, James **92–93**
 Huygens, Christiaan 78
 hydraulics 45
 hydrofoils 159
 hydrogen 94–96, 101, 103, 112, 189
 hydrogen balloons 68, 94
 hydrogen bomb 235, 239
 hydrogen bubble chamber 252
 hydrometers 32
 hygiene 26
 Hypatia **30–33**

I
 ice ages 117
 immune system 106–109, 147, 303
 immunology 303
 imprinting 231
 inertia 53
 infrared rays 99
 Ingenhousz, Jan **119**
 inheritance
 chromosomal 174
 genetics 140–143, 178–181
 inhomogenous solids 194

innate behavior 231
 insulin 250, 274–276
 internet 304–309
 isotopes 196, 239, 240, 252
 ivermectin 293
 IVF 261

J
 Jabir ibn Hayyan **39**
 Jacquard, Joseph 130
 Jefferson, Thomas 108
 Jemison, Mae Carol **311**
 Jenner, Edward **106–109**
 Jensen, Hans 241
 Jesty, Benjamin **109**
 Johnson, Katherine **310**
 Joule, James Prescott **138–139**
 Joule-Thomson Effect 139
 Julian, Percy **228**
 jumping genes 229
 Jupiter 52, 53

K
 K-electron capture 252
 Kandel, Eric **311**
 Kekulé, Friedrich August **161**
 Kelvin, Lord 139
 Kepler, Johannes **56–59**
 keratoprosthesis 297
 kinetic theory 150
 kingdoms 88
 Kirchhoff, Gustav 168
 kites 85
 Koch, Robert **146**
 Kornberg, Arthur 236

L
 Lamarck, Jean-Baptiste **119**
 land bridges 206
 Laplace, Pierre-Simon **119**
 Large Hadron Collider 280, 305
 lasers 239, 311
 Lavoisier, Antoine-Laurent de
 97, **100–103**
 Leakey, Louis 294
 learned behavior 231

Leavitt, Henrietta Swan **209**
 Leclerc, Georges-Louis **119**
 Lee, Tsung-Dao 253, 284
 Leeuwenhoek, Antoine van **73**
 Lehmann, Inge **217**
 Leibniz, Gottfried 32
 Lemaître, Georges 222
 lenses 72, 73, 164
 leprosy 225
 leukemia 286
 Levi-Montalcini, Rita **261**
 Licklider, J. C. R. **307**
 Liebig, Justus von **160**
 life, spontaneous generation of 146
 light
 absorption and emission of 168,
 170–171
 bending of 201
 color spectrum 77, 78
 and magnetism 123, 124, 148
 properties of 36
 scattering of 218
 speed of 150–151, 200
 in vacuums 68
 waves 78, 150–151, 218
 lightning 84–85
 Linnaeus, Carl **86–91**
 Lippershey, Hans **53**
 Liu Hui **38**
 lobotomies 193
 logarithms 127–128
 logic 14, 254, 256
 Lonsdale, Kathleen **260–261**
 Lorenz, Konrad **230–231**
 Lovelace, Ada 127, 130, **131**
 lunar eclipses 16
 Lyceum (Athens) 14, 17
 lysozyme 212

M
 magnetic fields 124
 magnetism 68, 124, 148, 150
 Mak, Tak Wah **303**
 malaria 26, 291–293
 Malpighi, Marcello **118**
 Malthus, Thomas 134
 mammoths 116–117
 Mandelbrot, Benoît **285**
 Manhattan Project 232, 235, 239,
 240, 253, 273, 279
 Marcus Aurelius 25, 26
 marine biology 178
 Mars 58
 mass

conservation of 101–102, 197
 energy and 201
 Mästlin, Michael 57
 mathematics
 al-Khwārizmī 34
 Archimedes 19
 Babbage 126–130
 Bernoulli, Jacob 118
 Brahmagupta 38–39
 da Vinci 43–45
 Euclid 38
 Fibonacci 44
 Fisher 209
 Fourier 119
 Herschel, John 129
 Hypatia 30–33
 Liu Hui 38
 Lovelace 131
 Mandelbrot 285
 Napier 63
 Noether 216
 Penrose 300
 Pythagoras 13
 Ramanujan 209
 Snell 63
 Thales 11
 Turing 254–259
 Wigner 240
 Zhang Heng 22
 matter 12, 186, 189, 201, 281, 300
 states of 200
 Maury, Matthew Fontaine **161**
 Maxwell, James Clerk 125,
 148–151, 196
 McClintock, Barbara **229**
 McCulloch, Ernst **286–289**
 mechanical energy 139
 mechanics
 Archimedes 19–20
 da Vinci 43
 Hypatia 30
 mechanization 127, 130
 medicine
 Avicenna 39
 Ball 225
 Banting 224
 Bath 297
 Berengario da Carpi 62
 Bernard 161
 Cajal 164–167
 Curie 185
 Edward 261
 Fallopio 62
 Finlay 161
 Fleming 212–215
 Florey 211
 Galen 24–29
 Golgi 167